

Hanfeng Huang

Shanghai, China · hhf13713678048@sjtu.edu.cn · +86 13713678048

Education

Shanghai Jiao Tong University

2021 Sep – 2025 Jun

- Bachelor of Science in Ship and Ocean Engineering – Mathematics and Applied Mathematics (Dual Degree)
- Advisor: **Tobias Diez**
- Core Math Courses: Mathematical Analysis (95/100), Linear Algebra (98/100), Lie Groups and Lie Algebras (96/100), Differential Geometry (98/100), Ordinary Differential Equations (95/100), Probability (98/100), Geometric Mechanics with Symmetries (98/100)

Ritsumeikan University

2026 Apr – Now

- Master of Science in Mathematics
- Advisor: **Daisuke Tarama**

Research Projects

Representation of $\mathfrak{su}(2)$ by complexification

2024 Jun

- Supervisor: **Tudor S. Ratiu**
- Semester report for the course Lie Groups and Lie Algebras
- Applied complexification technique to recover irreducible representations of $\mathfrak{su}(2)$ following Brian Hall's framework

Symplectic Reduction on Cotangent Bundle

2024 Dec

- Supervisor: **Tudor S. Ratiu**
- Semester report for the course Geometric Mechanics with Symmetries
- Studied symplectic reduction theorem and its application to cotangent bundles with symmetry groups

On the Deck Transformation

2024 Dec

- Supervisor: **Guanglian Zhang**
- Semester report for the course Algebraic Topology
- Analyzed deck transformations and covering space theory in the context of fundamental groups

Seminars

Reading Seminars: John M.Lee's *Introduction to Smooth Manifolds*

2024 Jan–Jun

- Supervisor: **Tobias Diez**
- Key topics: Smooth manifolds, tangent/cotangent bundles, vector fields, differential forms, Lie groups and Lie algebras, Maurer-Cartan forms

Reading Seminars: Loring W.Tu's *Differential Geometry*

2024 Sep–Nov

- Supervisor: **Tobias Diez**
- Key topics: Principal bundles, connections on vector bundles, Ehresmann connections, Theorema Egregium, affine connections

Reading Seminars: Shoshichi Kobayashi's *Differential Geometry of Connections and Gauge Theory*

2025 Sep–Dec

- Supervisor: **Tobias Diez**
- Key topics: Fiber bundles, connections on principal bundles, curvature forms, characteristic classes, Yang-Mills theory, gauge transformations

Summer School

- Summer school of Geometry in Utrecht University

2024 Aug

Awards and Scholarship

- Shanghai Jiao Tong University scholarship of level C (top 30% students only)
- WEICHAH Scholarship of Shanghai Jiao Tong University (top 30% students only)

2022 Dec

2023 Dec

Language Skills

- **Chinese:** Native
- **English:** IELTS 7, TOEFL 98
- **Japanese:** N2